

Deprecated Functions Identify Functions Procedure

Revision

Version 3
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SME

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Abstract

This document describes the procedure used to perform the function identification activity described in the **AVCDL** ^[1] secondary document **Currently Used Deprecated Functions Document** ^[2].

Group / Owner

Development / Software Developer

Motivation

This document is motivated by the need to document and justify the use of deprecated functions in the development of software for use within safety-critical, cyber-physical systems.

Note: Within the context of this document, the terms ***security*** and ***cybersecurity*** are used interchangeably. It is presumed that the term ***security*** is being used in reference to ***cybersecurity*** and not ***physical security***.

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Audience

The audience of this document is the cybersecurity SME who will be identifying deprecated functions.

Completeness of Output

Since the identification of deprecated functions is an as-is analysis, it is critical to ensure that the information gathered is as accurate and complete as possible. All information should be sourced from authoritative sources.

When information is not available for either a given section of the template or parts thereof, this should be noted. Major omissions should be recorded in the cybersecurity risk register.

Disposition of Output

Once completed, the generated output should be managed in the organization's document management system (DMS) as a document of record.

Entry Criteria

This document assumes that the reader understands the purpose of cybersecurity-relevant function deprecation. Further, that the reader has read and understood the **Currently Used Deprecated Functions Document** AVCDL secondary document.

Prerequisites – Security SME

Qualifications

It is required that the security SME is both qualified and trained in the handling of deprecated functions.

Note: The specific qualifications and training necessary to assert this expertise is outside the scope of this document.

Knowledge

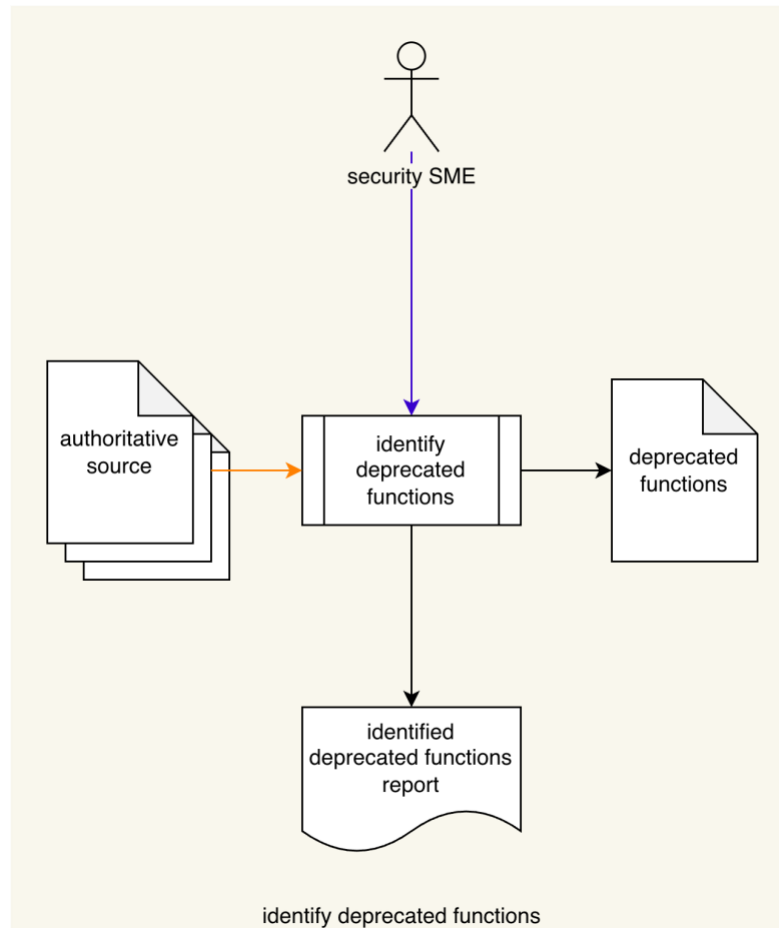
It is required that the security SME have expertise in the programming language under consideration whose deprecated functions are being reviewed.

Prerequisites – Input Materials

It is required that the cybersecurity SME acquire all relevant documentation related to the programming language under consideration necessary to complete the procedure.

Identify Functions Activity

The workflow diagram for the **identify deprecated functions** activity of the **deprecated functions** process is shown below.



The **Security SME** reviews the **Authoritative Sources** to identify functions which have been deprecated. These are entered into a **Deprecated Functions** list. An **Identified deprecated Functions Report** is generated.

Note: Deprecated functions should be considered on a per programming language basis.

The **Authoritative Source** material may include, but is not limited to:

- Technical standards
- Best practices
- Industry recommendations

Methodology

The methodology for this activity is straightforward.

All available documentation is reviewed and a list of deprecated functions is assembled.

Note: It is important to include all deprecated functions, even those which may not be applicable to specific products. Later activities will tailor the list to those which are deemed appropriate.

Deprecated Function Catalog Template

The deprecated functions may be documented using the **AVCDL deprecated function catalog template** Microsoft Excel workbook [\[3\]](#).

Note: Other forms of documentation are permissible so long as they provide the information laid out in this document.

There are four sheets in the workbook. They are:

- [Cover sheet](#)
- [Revision history](#)
- [Reference documents](#)
- [<Category>](#)

These will be addressed in turn.

Duplication of Rows in the Various Sheets

When there is the need to add rows to the various sheets of the workbook, be sure to duplicate an existing row. This is because validation checks may be attached to some of cells enabling the use of dropdown lists.

Cover Sheet

The **cover sheet** of the workbook is shown below:

Deprecated Functions									
Language	Language								
Platform	Platform								
Cybersecurity SME	Cybersecurity SME								
Date	1-Jan-2000								
Revision	1								

Fields to be completed are shown in **red**.

Language

This is the programming language under consideration.

Note: Only one language should be considered per file.

Platform

This is platform on which the language is used.

Note: In cases where the language has no platform specific considerations, “**any**” should be used.

Cybersecurity SME

This is the cybersecurity subject matter expert creating the list.

Date

This is the date when the list was created or updated. The date should be updated whenever the list is updated.

Revision

This is the revision number of the document. The revision number is a monotonic and increasing integer, starting at 1. It should be incremented every time the document is updated.

Revision History

The **revision history** sheet of the workbook is shown below:

Revision History		
Revision	Author	Description
1	cybersecurity SME	initial revision

Revision

The **revision** corresponds to that listed on the cover sheet.

Author

The **author** corresponds to the cybersecurity SME listed on the cover sheet.

Description

This is a brief description of changes made to the workbook since it was last updated.

Reference Documents

Note: These references are those necessary for the creation of the cybersecurity options list.

The **references** sheet of the workbook is shown below:

Reference Documents		
Name	Description	Location

Name

This is the name of the document being referenced.

Description

This is a brief description of the document being referenced.

Note: In cases where the reference material is expressed in the form other than a document (for example, a web page), this should be noted.

Location

This is the location of the document being referenced. It may be a physical location or a URL.

<Category>

The **category** sheet of the workbook is shown below:

Note: The name of the sheet and header should be replaced with the actual category of function. Some example categories are **memory**, **format**, **string**, and **time**.

<Category>					
ID	deprecated function	location	reason for deprecation	suggested replacement	notes

ID

This is the unique ID of the option.

Note: An ID of the form **DF<LANGUAGE>-<zero-leading number>** is recommended. For example: **DFCPP-05**.

Deprecated Function

This is the name of the function.

Location

This is the header containing the function.

Note: When the programming language under consideration does not utilize header files, the source file location should be used.

Reason for Deprecation

This field describes why the function has been deprecated.

Suggested Replacement

This is the replacement function or mechanism.

Notes

This is a general notes field.

Note: When multiple notes entries contain the same information, create a two column note reference section below the table (**note / information**), and giving a monotonically increasing number to each. Use that number in the **notes** field.

Identified Deprecated Functions Report

The **Identified Deprecated Functions Report** is the human-readable record of the activity. It is intended to be a standalone document for audit purposes. Its preferred embodiment for storage in the DMS is PDF. The content of the report is that captured in the abovementioned template.

Note: Report generation is covered more extensively in the **Report Generation** ^[4] AVCDL elaboration document.

Note: If possible, it is recommended that the report be generated programmatically.

Exit Criteria

This procedure is considered complete once the generated output has been entered into the organization's DMS as a document of record.

Note: The processes and procedures for entering documents into the DMS, or the updating thereof, are outside the scope of this document.

References

1. **AVCDL** (AVCDL primary document)
2. **Currently Used Deprecated Functions Document** (AVCDL secondary document)
3. **AVCDL deprecated function catalog template** (AVCDL template)
4. **Report Generation** (AVCDL elaboration document)